Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/18/2025
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	Hypobaric Plant Biology in Space Exploration - Molecular Responses of Arabidopsis to Combined Effects of Low Atmospheric Pressures and Microgravity of Spaceflight Vehicles		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	alp@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(352) 273-4855
Organization Name:	University of Florida		
PI Address 1:	Plant Molecular and Cellular Biology Program		
PI Address 2:	1301 Fifield Hall		
PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	10/21/2021	**End Date:**	10/22/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Haveman, Natasha	**Contact Phone:**	857-225-6244
Contact Email:	Natasha.J.Haveman@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	research scientist, Dr. Mingqi Zhou has left the project to pursue a faculty position. Abby Owens has joined the project as a lab manager and in technical support of this research		
COI Name (Institution):	Ferl, Robert Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC22K0214		
Performance Goal No.:			
Performance Goal Text:	NOTE: End date changed to 10/22/2025 per the PI (Ed., 2/14/25).		

Task Description:	Atmospheric pressure and composition are among the engineering variables considered in the design and construction of spaceflight vehicles and extraterrestrial habitats. Simply put, the costs of maintaining a pressure vessel at one atmosphere have been traded away throughout the history of spaceflight vehicle design and are traded away in future designs. And while the effects of hypobaria on plant biology are now well understood, there exist no studies on the combined effects of hypobaria and microgravity. In other words, the dominant physical manifestation of spaceflight, microgravity, has been left out of our understanding of plant hypobaria. We therefore propose to examine plant responses and physiological adaptations to the combined effects of low pressure while in the microgravity of the International Space Station (ISS). The objective of this proposal is to develop a refined understanding of the metabolic processes involved in plant responses and physiological adaptations to low pressure environments within space exploration vehicles and habitats. The long-term goal of this line of research is a fundamental understanding of low pressure plant biology within exploration vehicles and structures, with a practical goal of contributing to the design of plants that thrive in challenging exploration environments. The essential drivers of this project are that hypobaric environments will likely be a feature of future exploration vehicles and habitats, together with the knowledge that plants mount complex and costly metabolic responses to hypobaria. Furthermore, plants mount complex and sometimes unexpected responses to spaceflight and altered gravity environments. We hypothesize that the combination of hypobaria and microgravity will have a synergistic effect on the physiological adaptation to this complex environment, and that the patterns of gene expression will reveal strategies to both understand and help ameliorate the combined effect. These responses will also inform fundamental understanding of how plants adapt to changing terrestrial habitats facing complex and novel stressful environments. There is a facility on the ISS that is well suited for these experiments: the Combustion Integrated Rack (CIR). The CIR has a pressure vessel that can be programmed telemetrically to the gas composition and pressures relevant to our study. There is also a CIR ground unit that can be similarly programmed for parallel ground controls, thereby enabling the precise dissection of the effects of the orbital environment on plant biology within the pressure vessel. The proposed research will provide fundamental insights into the biological impact of novel atmospheric environments, a focus area that is itself specifically identified in the Decadal Study. (Ed. Note. The National Academies of Sciences, Engineering, and Medicine Decadal Survey on Biological and Physical Sciences Research in Space 2023-2032). More importantly, this Hypobaric Plant Biology in Space Exploration study seeks to fundamentally examine combined spaceflight effects in order to develop an understanding of emergent response properties that are not predicable from an examination of the individual responses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research contributes to a greater understanding of the role of transcriptomic changes to the plant in response to both hypobaric and microgravity environments. Further, it will be the first experiment to test whether the interplay between hypobaric stress to plants and microgravity growth of plants has synergistic effects.
Task Progress:	There are very few updates since the 2024 annual report as the hardware proposed for this experiment, the Combustion Integrated Rack (CIR) hardware, on board the International Space Station (ISS), is no longer available, and so progression to Science Verification Testing (SVT) stalled while new hardware options are under investigation. The NASA Kennedy Space Center (KSC) Team (led by Lucy Orozco and Natasha Haveman) propose using the BRIC-100VC hardware to create the hypobaric environments required for the APEX-11 experiment. [Ed. Note: APEX is a short name for NASA's Advanced Plant Experiments.] The Biological Research in Canister (BRIC) is an anodized-aluminum cylinder used to provide passive stowage for investigations of the effects of spaceflight on small specimens. The BRIC 100 mm petri dish vacuum containment unit (BRIC-100VC), which has previously supported a plant somatic embryo development experiment (Hemerocallis lilioasphodelus), is required to hold a 5% CO2 gaseous environment for 4.5 months in space (Levine et al., Adv Space Res, 2000). The BRIC-100VC is a completely sealed, anodized-aluminum cylinder providing containment and structural support of the experimental specimens. The top and bottom lids of the canister include rapid disconnect valves for filling the canister with selected gases. These specialized valves allow for specific atmospheric containment within the canister, providing a gaseous environment defined by the investigator. Additionally, the top lid has been designed with a toggle latch and O-ring assembly, allowing for prompt sealing and removal of the lid. The outside dimensions of the BRIC-100VC canisters are 16.0 cm (height) x 11.4 cm (outside diameter). The lower portion of the canister has been equipped with sufficient storage space for passive temperature and relative humidity data loggers. The BRIC- 100VC canister has been optimized to accommodate standard 100 mm laboratory petri dishes or 50 mL conical tubes. Depending on storage orientation, up to 6 or 9 canisters have been flown within an ISS stowage locker. The adaptation of the passive BRIC-100VC hardware will require an experiment service panel, the schematic for which is currently under development by a flight design engineer. The idea is that the service panel will seamlessly connect to the Vacuum Exhaust System (VES) through a hose and quick disconnect (QD) on one end, while establishing a link to the O2/N2 tank on the other end. Designed with precision, it will incorporate essential components, such as a filter, a vent, pressure and oxygen gauges, flexible hoses, QD's, and both ball and needle valves, ensuring optimal performance and reliability for the experiment operations. While the hardware is under development, we are taking the opportunity to further calibrate our green fluorescent protein (GFP) reporter gene lines and work with the imaging capabilities of the Keyence microscope, which will be used for imaging on the Flight and Ground experiments. REFERENCES Levine HG, Anderson KF, Krikorian AD. "The 'gaseous' environment in sealed BRIC-100VC canisters flown on 'Mir'with embryogenic daylily cell cultures." Advances in Space Research. 2000 Jan 1;26(2):307-10. https:// ; PMID: 11543167
Bibliography Type:	Description: (Last Updated: 11/20/2025)